

BUS33-1

Advanced Marketing

Growth & Customer Analytics

Marketing as allocation decisions defended with evidence.

Albert School — 22 hours

Contents

1	The AARRR growth framework	2
2	Acquisition-channel economics	2
2.1	Channel families	3
2.2	Viability condition	3
3	Growth loops vs funnels	3
4	A/B testing and experimentation discipline	4
4.1	Pre-committed decision rule	4
4.2	Statistical power	4
4.3	Significance vs effect size	4
4.4	Conversion-funnel drop-off diagnosis	4
5	Multi-touch attribution	4
6	Customer analytics	5
6.1	Customer Lifetime Value	5
6.2	RFM segmentation	5
6.3	Cohort analysis	5
6.4	Churn early-warning signals	5
7	Performance vs brand, and marketing automation	6
7.1	The performance–brand tension	6
7.2	Marketing automation	6
8	Consumer anthropology	6
8.1	Theories of consumption	6
8.2	Jobs-to-be-Done (Christensen)	6
8.3	Diffusion of innovation (Rogers)	6
9	Integrated growth plan	7

1 The AARRR growth framework

AARRR (also called “pirate metrics”) models a customer’s relationship with a product as five stages. Each stage is measured by a conversion rate from the previous stage.

Definition 1 (AARRR) The five stages are:

- **Acquisition** — a prospect first reaches the product (visits the site, installs the app).
- **Activation** — the prospect has a first satisfactory experience (the “aha moment”).
- **Retention** — the user returns and keeps using the product over time.
- **Revenue** — the user performs a monetising action (purchase, subscription).
- **Referral** — the user brings new prospects to the product.

Applying AARRR to a company means defining each stage operationally for that company, measuring the conversion rate at each stage, and locating the stage with the largest drop-off as the priority for action. Acquisition channels are then selected on channel-economic logic (Chapter 2) rather than on channel fashion.

2 Acquisition-channel economics

Each acquisition channel has a cost and an output. The governing quantity is the cost of acquiring one customer.

Definition 2 (CAC) The **Customer Acquisition Cost** of a channel over a period is

$$\text{CAC} = \frac{\text{total spend on the channel}}{\text{number of customers acquired through it}}.$$

2.1 Channel families

- **Paid** — Google Ads (search intent), Meta (interest/behaviour targeting). Spend buys impressions or clicks; CAC is directly observable and scales with budget, but rises as the channel saturates.
- **Organic** — SEO and content. No per-click cost, but a fixed production and time cost; CAC falls as content compounds traffic.
- **Social** — non-paid distribution on social platforms; reach depends on sharing and platform algorithms.
- **Partnerships** — distribution through another company's audience, priced as a fixed fee, a revenue share, or an affiliate commission.

2.2 Viability condition

A channel is viable for a business only if the value of a customer exceeds the cost of acquiring one through that channel:

Proposition 3 (Channel viability) A channel is viable when

$$\text{LTV} > \text{CAC},$$

where LTV is the customer lifetime value (Chapter 6). A common benchmark for a healthy business is $\frac{\text{LTV}}{\text{CAC}} \geq 3$, with CAC recovered (paid back) within 12 months.

CAC benchmarks differ by channel and by business type: a high-ACV B2B business can sustain a far higher CAC than a low-price-point B2C business, because its LTV is higher. Viability is therefore judged per channel and per business type, not against a universal CAC figure.

3 Growth loops vs funnels

A **funnel** is linear: input flows in at the top, a fixed fraction converts at each stage, and the output leaves at the bottom. To grow a funnel, you must keep buying more input — it is a “leaky bucket.”

Definition 4 (Growth loop) A **growth loop** is a mechanism in which the output of one cycle becomes the input of the next, so that growth compounds. Example structure: existing users produce something (referrals, content, data) that acquires new users, who in turn produce more of the same.

The two frameworks apply in different situations:

- A **funnel** describes a one-way conversion process and is the right framework for diagnosing where prospects drop off.
- A **growth loop** describes a self-reinforcing acquisition mechanism and is the right framework when acquisition is driven by the existing user base.

A loop **compounds** only while each cycle feeds the next; it becomes a leaky funnel once a **constraint** (market saturation, falling referral rate, rising cost) breaks the feedback. Building

a growth-loop model for a company means identifying that constraint — the variable that will stop the loop at scale.

4 A/B testing and experimentation discipline

An A/B test compares two variants (A = control, B = treatment) by randomly assigning users to each and measuring a chosen metric.

4.1 Pre-committed decision rule

Before running the test, fix: the hypothesis, the single primary metric, the minimum effect size worth detecting, the sample size, and the rule that maps the result to a decision. Committing to these in advance prevents reading a result after the fact to fit a preferred conclusion.

4.2 Statistical power

Definition 5 (Significance and power)

- The **significance level** α is the probability of declaring a difference when none exists (false positive).
- The **power** $1 - \beta$ is the probability of detecting a real difference of the assumed effect size (true positive); β is the false-negative probability.

Power depends on α , the effect size, the baseline conversion rate, and the sample size. Fixing α (typically 0.05) and a target power (typically 0.80) together with the minimum detectable effect determines the required sample size per variant, which must be computed **before** the test so the test runs to that size.

4.3 Significance vs effect size

Statistical significance states only that an observed difference is unlikely under the no-difference hypothesis; it does not state that the difference is large. With a large enough sample, a trivial difference becomes significant. A read must therefore report the **effect size** (how large the difference is) alongside significance, and judge it against the minimum effect worth acting on.

4.4 Conversion-funnel drop-off diagnosis

To diagnose a funnel, measure the conversion rate between each pair of consecutive stages. The stage with the steepest drop-off is the binding constraint and the target for the next experiment.

5 Multi-touch attribution

A customer typically touches several channels before converting. Attribution assigns credit for the conversion across those touchpoints.

Definition 6 (Attribution models)

- **Last-click** — assigns 100% of the credit to the final touchpoint.
- **Linear** — splits the credit equally across all touchpoints.
- **Data-driven** — assigns credit using a model estimated from the data (e.g. comparing conversion with and without each touchpoint).

Each model carries a bias:

- Last-click over-credits closing channels (e.g. branded search) and under-credits channels that create demand earlier in the journey.

- Linear assumes every touch contributes equally, which is rarely true.
- Data-driven removes the fixed-weight assumption but depends on data volume, on the quality of the underlying model, and on touchpoints being observable.

Because the chosen model changes which channels appear efficient, it changes the budget-allocation decision. Selecting a model means matching it to the business context, and reasoning explicitly about how its bias would shift allocation — i.e. reasoning under attribution ambiguity rather than treating any single model’s output as ground truth.

6 Customer analytics

6.1 Customer Lifetime Value

Definition 7 (CLV) The **Customer Lifetime Value** is the total margin a customer is expected to generate over the relationship. A simple form is

$$\text{CLV} = \frac{\text{ARPU} \times \text{gross margin}}{\text{churn rate}},$$

where ARPU is average revenue per user per period and the churn rate is the fraction of customers lost per period. Future cash flows may be discounted.

6.2 RFM segmentation

Definition 8 (RFM) **RFM segmentation** scores each customer on three dimensions:

- **Recency** — how recently they last purchased,
- **Frequency** — how often they purchase,
- **Monetary** — how much they spend.

Customers are ranked (e.g. into quintiles) on each dimension; the combined score defines segments (e.g. high-value loyal, at-risk, lapsed).

6.3 Cohort analysis

Definition 9 (Cohort analysis) A **cohort** is a group of customers defined by a shared start period (e.g. all customers acquired in a given month). Cohort analysis tracks a metric (retention, revenue) for each cohort over the periods since acquisition, isolating behaviour by acquisition date rather than blending all customers together.

Cohort analysis is typically run on real transaction data, grouping transactions by acquisition period and the number of periods elapsed.

6.4 Churn early-warning signals

Churn is the loss of a customer. Early-warning signals are observable changes that precede churn — falling usage frequency, lengthening time since last purchase (rising recency in RFM terms), declining order value, or reduced engagement. Detecting them allows intervention before the customer is lost.

7 Performance vs brand, and marketing automation

7.1 The performance–brand tension

- **Performance marketing** targets measurable short-term response (clicks, conversions) attributable to specific spend.
- **Brand marketing** builds long-term mental availability and is not directly attributable to a single conversion.

The tension is an allocation trade-off: performance is measurable and immediate but does not compound demand, while brand compounds demand but resists short-term measurement. Optimising only what is measurable systematically underfunds brand.

7.2 Marketing automation

Marketing automation tools — CRM, segmentation engines, lifecycle/email campaigns — execute marketing actions at scale and on triggers. They are an **execution tool**: they implement segmentation and lifecycle decisions efficiently but do not, by themselves, decide which segments to serve or how to position. Automation executes a strategy; it does not constitute one.

8 Consumer anthropology

This layer explains **why** consumers choose, distinguishing durable structural trends from passing fads.

8.1 Theories of consumption

- **Utility** — consumption as the satisfaction of functional needs.
- **Social distinction (Veblen)** — **conspicuous consumption**: goods consumed to signal status, where high price is itself part of the appeal.
- **Distinction (Bourdieu)** — taste as a marker of social position; consumption reproduces cultural and social distinctions between groups.

8.2 Jobs-to-be-Done (Christensen)

Definition 10 (Jobs-to-be-Done) Customers “hire” a product to do a **job** — to make progress in a particular circumstance. The unit of analysis is the job, not the customer demographic or the product category. A product competes with every alternative the customer could hire for the same job.

JTBD identifies the underlying job a category serves, which reveals genuine competitors and distinguishes a durable need from a particular product form.

8.3 Diffusion of innovation (Rogers)

Definition 11 (Diffusion of innovation) An innovation spreads through a population in sequence across adopter categories: **innovators**, **early adopters**, **early majority**, **late majority**, and **laggards**. The cumulative adoption curve is S-shaped.

Diffusion logic distinguishes a **structural trend** — which follows the adoption curve through successive categories — from a **fad**, which spikes among early adopters and collapses without crossing into the majority. Applied with JTBD to a real category, it separates genuine trends from fashions.

9 Integrated growth plan

The course culminates in a complete growth plan for a real company, assembling the preceding chapters into one allocation decision:

- **Channel selection** — choose acquisition channels on channel economics and AARRR conversion benchmarks (Chapters 1–2).
- **Budget allocation** — distribute spend across channels under explicit attribution assumptions, with a **sensitivity analysis** across the plausible range of those assumptions (Chapter 5).
- **KPI dashboard** — the AARRR-stage metrics, CAC by channel, and the customer- analytics measures (CLV, retention, churn signals) used to monitor the plan (Chapters 1, 6).
- **12-month CAC/LTV projection** — a forward projection of CAC by channel and LTV, used to defend the recommendation against observable business constraints (Chapters 2, 6).

The plan is defended by stating the assumptions, the evidence behind each quantified figure, and how the recommendation changes as assumptions vary.